

Unit V

Environmental Policy, Legislation and EIA

Environmental Protection Act, 1986

The law was enacted after the Bhopal gas tragedy, this Act gives the Central Government broad powers to protect the environment and manage pollution.

Objectives:

- Prevent environmental pollution.
- Empower central authorities to act.
- Implement global environmental agreements.

Highlights:

- Covers pollution of air, water, and land.
- Allows the government to:
 - Set pollution control norms.
 - Regulate industries and hazardous substances.
 - Act during environmental emergencies.
- Officials can inspect sites and enforce compliance.

Penalties:

- Up to 5 years imprisonment or a fine up to ₹1, 00,000.
- Continued offenses face stricter penalties.

Importance:

- Forms the base for various environmental rules like waste management and EIA.
- Aids in protecting health and ecosystems.

Limitations:

- Enforcement gaps, limited public participation, and overlapping laws affect effectiveness.

Air (Prevention and Control of Pollution) Act, 1981

The Act was enacted following the 1972 Stockholm Conference to specifically address air pollution and protect both environmental and public health.

Objectives:

- Prevent and reduce air pollution.
- Establish Central and State Pollution Control Boards (CPCB & SPCBs).
- Ensure that industries comply with air quality regulations.

Key Provisions

- Defines “air pollutant” as any harmful substance in the atmosphere.
- Bans polluting fuels and substances in designated control zones.
- Industries must obtain “Consent to Operate” from SPCBs.
- CPCB and SPCBs are empowered to:
 - Monitor air quality.
 - Set and enforce emission standards.
 - Advise governments and penalize violations.

Penalties

- Non-compliance can result in fines, imprisonment, or even industrial closure significance
- Critical for managing industrial emissions.
- Promotes cleaner air to safeguard human health and the ecosystem.

Limitations

- Weak enforcement in many regions.
- Inadequate monitoring infrastructure.

Water (Prevention and Control of Pollution) Act, 1974

This was India's first major environmental legislation, enacted to tackle water pollution and improve the quality of rivers, lakes, and water bodies.

Objectives

- Prevent and control water pollution.
- Restore and maintain water quality.
- Establish Central and State Pollution Control Boards (CPCB & SPCBs) to oversee enforcement.

Key Provisions

- Defines pollution as any contamination harmful to health, safety, or the environment.
- Discharge of sewage or industrial waste into water bodies is prohibited without prior consent.
- CPCB and SPCBs are authorized to:
 - Monitor water quality.
 - Set standards for wastewater discharge.
 - Inspect industries and collect samples.
 - Initiate legal action against polluters.

Penalties

- Violations can lead to imprisonment up to 3 months and/or a fine up to ₹10,000.
- Continued non-compliance may result in additional daily fines.

Significance

- First law to address water pollution in a legal framework.
- Promotes wastewater treatment before disposal.
- Foundation for India's water quality management efforts.

Limitations

- Inadequate resources and manpower in Pollution Control Boards.

- Weak enforcement in several states.
- Low public awareness and limited participation.
- Limited public awareness and engagement.

Forest (Conservation) Act, 1980

The law was enacted in response to rapid deforestation and ecological damage, this act aims to preserve India's forest resources and ensure their sustainable use.

Objectives

- Conserve forests and prevent their unauthorized exploitation.
- Regulate non-forest use of forest land (e.g., for mining, industry, roads).
- Maintain ecological balance and protect biodiversity.

Key Provisions

- Central Government approval is mandatory before using forest land for non-forest purposes.
- Applies to both government and private forests.
- Includes:
 - Compensatory afforestation (replanting trees when forests are diverted).
 - A formal project clearance process for forest land use.
- The Ministry of Environment, Forest and Climate Change (MoEFCC) oversees implementation.

Amendments

- The 1988 amendment enhanced forest protection.
- Recent revisions aim to balance development and conservation, though subject to strict conditions.

Importance

- Protects forest cover and wildlife habitats.
- Discourages illegal deforestation.

- Encourages sustainable land-use practices.

Limitations

- Delays in project approvals can slow legitimate development.
- Enforcement weaknesses may still allow degradation.
- Conflicts may arise with tribal and local communities over land rights.

Wildlife (Protection) Act, 1972

This was one of India's earliest comprehensive laws to protect wildlife and biodiversity, addressing poaching, illegal trade, and habitat loss.

Objectives

- Conserve wild animals, birds, and plants.
- Control hunting, poaching, and illegal wildlife trade.
- Establish protected areas like National Parks and Wildlife Sanctuaries.

Key Provisions

- Hunting of endangered species is strictly prohibited.
- Wildlife species are categorized into six schedules:
 - Schedule I & II: Complete protection; severe penalties (e.g., tiger, elephant).
 - Schedule III & IV: Protected with lesser penalties.
 - Schedule V: Species that may be hunted (e.g., certain vermin).
 - Schedule VI: Protects rare and threatened plant species.
- Establishes:
 - Wildlife Sanctuaries
 - National Parks
 - Reserved and Closed Areas
- Regulates:
 - Trade, possession, and transport of animal and plant parts.

Authorities

- Wildlife Advisory Boards (Central and State).

- Forest officers and wildlife wardens responsible for enforcement.

Penalties

- Harsh punishments for offenses involving Schedule I species:
 - Up to 7 years' imprisonment and heavy fines.

Importance

- Crucial for the protection of endangered species and habitats.
- Facilitates the creation and management of protected areas.

Limitations

- Doesn't fully address human-wildlife conflicts.
- Poaching and illegal trade still occur.
- Limited staff and funding hamper enforcement.

Municipal Solid Waste (MSW)

MSW Rules, 2000

- Introduced by: Ministry of Environment and Forests
- Purpose: To ensure cities and towns manage their solid waste responsibly.
- Applies to: Urban local bodies (municipalities)

Main Guidelines:

- Waste should be separated into three categories: biodegradable, recyclable, and inert.
- Different colored bins should be used to store each type of waste.
- Waste collection must happen daily and be transported in closed vehicles.
- Composting and recycling methods should be encouraged.
- Only non-recyclable waste should go to landfills.
- Municipalities must report their waste management activities every year.

Solid Waste Management (SWM) Rules, 2016:

- **Replaces:** MSW Rules, 2000

- **Expanded Scope:** Not just municipalities but offices, housing societies, industries, etc., also included.

Key Features:

- Waste segregation at the source is compulsory, into:
 - Biodegradable waste
 - Dry (recyclable) waste
 - Domestic hazardous waste
- Citizens must pay a user fee for waste collection; fines are imposed for littering.
- Burning of waste is completely banned.
- Large waste generators (more than 100 kg/day) must treat their waste on-site.
- Waste pickers are to be included in the formal waste management system.
- Local-level waste processing (like composting) is encouraged.

Comparison: MSW 2000 vs. SWM 2016

Feature	MSW Rules, 2000	SWM Rules, 2016
Focus	Municipal authorities	All waste generators
Segregation	Suggested	Made compulsory
Informal sector	Not mentioned	Included formally
Waste burning	Not clear	Explicitly banned
User charges	Not specified	Introduced

Conclusion:

The 2016 Rules improved waste management by being more inclusive and environmentally responsible. They encourage people and institutions to take active roles in managing waste.

Biomedical Waste (BMW) Management Rules, 2016

(Replaces BMW Rules, 1998)

- **Issued by:** Ministry of Environment, Forest and Climate Change (MoEFCC)
- **Objective:** To make biomedical waste handling stricter, more practical, and eco-friendly.

Applicability extended to

- Clinics, blood banks, veterinary institutions, research labs, and more.

Key Provisions:

- Waste must be sorted at the source into 4 color-coded categories:
 - **Yellow:** Human and animal waste, soiled items – must be incinerated.
 - **Red:** Contaminated plastics – treated using autoclaving or shredding.
 - **White (translucent):** Sharps like needles – disinfected and sealed.
 - **Blue:** Glassware – disinfected and recycled.
- Lab waste must be treated before disposal.
- A **barcode system** is introduced to track biomedical waste.
- All healthcare facilities must get permission from the State Pollution Control Board (SPCB) and renew it regularly.
- Training for staff, public awareness, and keeping records are mandatory.

Comparison: BMW 1998 vs BMW 2016

Feature	BMW Rules, 1998	BMW Rules, 2016
Applicability	Hospitals only	All healthcare-related units
Waste Categories	10 categories	4 simplified categories
Waste Tracking	Not present	Barcode-based system
Waste Pre-treatment	Optional	Mandatory in some cases
Training/Reporting	Not mandatory	Made compulsory

Conclusion:

These rules aim to reduce infection risks and pollution by ensuring proper disposal, segregation, and tracking of biomedical waste.

Hazardous and Other Wastes Rules, 2016

- **Issued by:** Ministry of Environment, Forest and Climate Change (MOEFCC)
- **Scope:** Covers both hazardous wastes and other wastes like e-waste, used oil, old batteries, etc.

Key Highlights:

- Hazardous waste is defined as anything that is toxic, flammable, reactive, corrosive, or explosive.
- **Responsibilities of Waste Generators:**
 - Reduce waste generation.
 - Maintain proper documentation.
 - Store, label, and dispose of waste correctly.
- Authorization is required from the SPCB to handle such waste.
- Manifest System introduced: Tracks waste from origin to final disposal.
- Import/export of hazardous waste is allowed only for recycling, not for dumping.
- TSDFs (Treatment, Storage, and Disposal Facilities) must be used for final disposal.

Comparison: Hazardous Waste Rules 1989 vs 2016

Feature	1989 Rules	2016 Rules
Scope	Hazardous waste only	Includes other specific wastes too
Focus	Waste disposal	Recycling and resource recovery
Waste tracking	Not included	Manifest system introduced
International transport	Limited	Clearly defined and regulated

Conclusion:

The 2016 Rules emphasize safer and more sustainable waste management. They support recycling, stricter controls on hazardous materials, and ensure safer handling and disposal practices.

Environmental Impact Assessment (EIA)

Structure of an EIA Report

An EIA report follows a standard format set by authorities like the MoEFCC in India. It includes:

Introduction

- States the purpose and scope of the EIA.

- Gives background about the project.

Project Details

- Information on project location, size, layout, processes, and resource needs (materials, water, and energy).
- Covers both construction and operation stages.

Existing Environmental Conditions (Baseline Study)

- Describes the current environmental status in the area (10–15 km radius).
- Covers air, water, soil, noise, land use, ecology, and socio-economic conditions.

Expected Environmental Impacts & Control Measures

- Identifies possible negative effects during construction and operation.
- Suggests actions to reduce or prevent these impacts.

Analysis of Alternatives

- Compares different options (sites, technologies, processes) to choose the best one.

Environmental Management Plan (EMP)

- Outlines how mitigation measures will be applied.
- Includes monitoring schedules and reporting procedures.

Risk and Disaster Planning

- Lists safety precautions and emergency plans for accidents or hazards.

Public Participation

- Summarizes public opinions and how concerns were addressed.

Summary and Recommendations

- Concludes with the overall environmental acceptability of the project.

Methods of Collecting Baseline Data

Baseline data refers to environmental information collected before the project starts. It helps in predicting changes and impacts.

Parameter	How Data Is Collected
Air Quality	24-hour air monitoring (PM ₁₀ , PM _{2.5} , SO ₂ , NO ₂) using air samplers
Water Quality	Water sampling and testing for pH, BOD, COD, heavy metals
Noise Levels	Sound meters measure continuous noise levels (Leq)
Soil Quality	Soil samples tested for nutrients, pH, metals
Flora and Fauna	Field surveys, transect walks, data from Forest Department
Land Use	Satellite images, GIS maps, and on-ground surveys
Socio-Economic Conditions	Household surveys, group discussions, census data
Hydrology/Geology	Field visits, geological and groundwater board reports

Conclusion:

A proper EIA report and accurate baseline data help plan sustainable development by predicting and controlling environmental impacts effectively.

EIA Focus: Impact on Air

Air Pollution Sources in Projects

Many large-scale projects like industries, roads, and mining can harm air quality. Major pollution sources include:

- Smoke from fuel burning (CO₂, NO_x, SO₂, PM)
- Dust from construction and mining
- VOCs from chemical use
- Gas leaks from storage or pipelines

Major Air Pollutants

- Particulate Matter (PM_{2.5}, PM₁₀)

- Carbon Monoxide (CO)
- Sulphur Dioxide (SO₂)
- Nitrogen Oxides (NO_x)
- Ozone (O₃)
- Volatile Organic Compounds (VOCs)
- Greenhouse Gases (GHGs)

How EIA Assesses Air Quality

- Baseline Study: Checks current air quality in the project area
- Emission Inventory: Lists and estimates emissions from project activities
- Dispersion Modelling: Predicts how pollutants will spread in the air
- Impact Prediction: Compares air quality with and without the project

Measures to Reduce Air Pollution

- Install filters and scrubbers to clean emissions
- Plant trees around the site (green belt)
- Spray water to control dust
- Use clean fuels and efficient machines
- Monitor air quality regularly

Laws Related to Air Impact in India

- Air Act, 1981
- Environment Protection Act, 1986
- National Ambient Air Quality Standards (NAAQS)
- EIA Notification, 2006

Why Air Impact Assessment Is Important

- Protects human health from diseases
- Reduces smog and acid rain
- Helps in legal compliance
- Supports clean and sustainable growth

EIA Focus: Impact on Water

Water Pollution from Projects

Common sources of water pollution in development projects:

- Industrial waste discharged into rivers or seas
- Sewage from cities and construction sites
- Surface runoff from mining
- Excess use of groundwater
- Heat pollution from power plants
- Oil or chemical leaks

Types of Water-Related Impacts

- Contamination of water (chemical, biological, thermal)
- Reduced flow or blocked drainage
- Falling groundwater levels
- Algae growth (eutrophication) due to waste nutrients
- Harm to aquatic life
- Risk of floods

Water Quality Assessment in EIA

- Baseline Study: Tests water quality before the project begins
- Sampling & Testing: Checks for pH, BOD, COD, metals, bacteria
- Hydrological Models: Predict water flow changes and flood risks
- Impact Prediction: Anticipates how the project will affect water

Water Pollution Control Methods

- **ETPs** for industrial waste
- **STPs** for domestic waste
- Rainwater harvesting and recharge pits
- **Zero Liquid Discharge (ZLD)** systems
- Storm water drains and sediment traps

- Keeping buffer zones around water bodies

Indian Water Laws & Standards

- Water Act, 1974
- Environment Protection Act, 1986
- EIA Notification, 2006
- CGWA Guidelines for groundwater use
- IS:10500 for drinking water quality

Importance of Water Impact Studies

- Ensures clean drinking water
- Protects rivers, lakes, and aquatic life
- Supports agriculture and rural livelihoods
- Helps follow environmental laws
- Encourages sustainable water use

EIA: Biological and Socio-Economic Aspects

(Essential summary – paraphrased section continued from your input)

EIA doesn't only consider air, water, and soil, but also:

- **Biological environment:** Includes plant and animal life, biodiversity, and protected species.
- **Socio-economic environment:** Considers the well-being of local communities, livelihoods, health, land use, and cultural aspects.

These are crucial to make sure development is:

- **Environmentally safe** (protecting species and ecosystems), and
- **Socially fair** (benefitting communities, minimizing displacement and inequality).

Biological Environment: The concept deals with

- Types and number of plant and animal species.

- Habitats like forests, wetlands, grasslands, and mangroves.
- Rare, endangered, and native species.
- Wildlife migration routes and breeding areas.
- Protected areas (National Parks, Sanctuaries, and Biosphere Reserves).

Possible Impacts

- Habitat loss due to land use changes.
- Decrease in biodiversity.
- Ecosystems breaking into smaller patch.
- Pollution harming plants and animals.
- Invasive species disrupting local ecosystems.

How it's studied

- Field surveys to record plants and animals.
- Satellite images and GIS mapping for habitat study.
- Biodiversity scoring methods.
- Predictive tools like habitat modelling.

Preventive Measures

- Plans to protect endangered species.
- Planting trees and creating green spaces.
- Restoring damaged habitats.
- Avoiding ecologically sensitive areas.
- Replanting forests as per legal requirements.

Socio-Economic Environment

Deals with

- Population, jobs, and land use.
- Farming, fishing, industries.
- Health, education, and sanitation.
- Cultural/religious sites and local services.

Possible Impacts

- People losing homes or land.
- Job loss and livelihood issues.
- Overuse of local resources like water and firewood.
- Health problems from pollution.
- Damage to local traditions or heritage.

How it's studied

- Surveys and interviews with affected people.
- Meetings with the community and officials.
- Economic and social impact analysis.
- GIS mapping of affected areas.

Preventive Measures

- Resettlement and compensation plans.
- Job training and support programs.
- Setting up health and sanitation facilities.
- Creating jobs through project-linked work.
- Investing in local infrastructure and services.

Legal Framework (India)

- Wildlife (Protection) Act, 1972
- Forest (Conservation) Act, 1980
- Biological Diversity Act, 2002
- Forest Rights Act, 2006
- Land Acquisition and R&R Act, 2013
- EIA Notification, 2006

Importance

- Maintains ecological balance and species protection.
- Safeguards communities and their rights.

- Ensures development is fair and sustainable.
- Promotes transparency and public input.
- Helps in resolving conflicts before they arise.

Risk Assessment in EIA

Risk Assessment helps identify possible dangers from a project and plan how to avoid or manage them.

1. Hazard Identification

- Find out what can go wrong using field data, expert input, and checklists.

2. Dose–Response Assessment

- Study how much exposure to a substance causes harm, using lab or health data.

3. Exposure Assessment

- Measure how and how often people or nature come into contact with risks.

4. Risk Characterization

- Estimate the severity and chance of negative outcomes (low, medium, high).

5. Risk Evaluation

- Compare risk levels with legal limits or standards.

6. Managing Risks

- Use pollution controls, safety rules, emergency plans, and smart land use.

7. Handling Uncertainty

- Check how data errors or unknowns could affect risk estimates using simulations.

8. Sharing with Stakeholders

- Communicate risks clearly with communities, officials, and developers.

Tools Used

- HAZOP, FMEA, QRA, GIS mapping, Fault Tree and Event Tree Analysis.

Importance

- Prevents accidents and protects health and nature.
- Helps in getting legal approvals.
- Builds trust and prepares for emergencies.

Sustainable Development

Development that meets present needs without harming the ability of future generations to meet their needs.

Three Pillars

1. **Environmental** – Save resources, reduce pollution, protect biodiversity.
2. **Economic** – Inclusive growth, efficient use of resources, clean energy.
3. **Social** – Fairness, access to education/health, respect for culture.

Key Principles

- **Intergenerational equity:** Fairness across generations.
- **Precautionary principle:** Avoid unknown risks.
- **Polluter pays:** Those causing pollution must fix it.
- **Public participation:** People must be involved.
- **Carrying capacity:** Don't exceed nature's limits.

UN Sustainable Development Goals (SDGs)

Global targets (by 2030) include:

- No poverty
- Quality education
- Clean energy

- Climate action
- Sustainable cities
- Life below water and on land

Role of EIA

- Minimizes environmental and social damage.
- Helps in smart, long-term planning.
- Ensures responsible and lawful development.

Conclusion

EIA helps in building a future where economic growth, environmental care, and social justice go hand in hand. It is key to achieving sustainable development.

Population and Its Explosion

Population explosion means a very fast and uncontrolled increase in the number of people, especially in developing countries. This happens because the number of births stays high while the number of deaths goes down due to better healthcare and technology.

Explosion reasons of population

- Better health services have reduced deaths.
- High birth and fertility rates continue in many areas.
- Lack of awareness about family planning methods.
- Cultural and religious reasons favour big families.
- Early marriages lead to longer childbearing years.
- People moving to cities in search of jobs or better life.

Effects of Population Explosion

Environmental Effects

- Overuse of natural resources like water, land, and minerals.
- Cutting down forests, harming animals and plants.
- Water shortages and pollution.

- More air and soil pollution.
- Global warming and climate change.

Social and Economic Effects

- More poverty and joblessness.
- Not enough houses, roads, and basic facilities.
- Pressure on hospitals and schools.
- Not enough food for everyone.
- Crowded cities and growing slums.

Methods to Control Population Growth

Spread Awareness and Education

- Make education available for everyone, especially girls.
- Teach about family planning and reproductive health.
- Promote later marriages and proper spacing between children.

Improve Healthcare

- Provide cheap and easy access to contraceptives and maternal care.
- Run family welfare programs.
- Offer safe abortion options (where legal).

Government Action

India's National Population Policy (2000)

- Goal: Stable population by 2045.
- Bring fertility rates to replacement levels.
- Make family planning services easily available.

Economic and Social Support

- Give benefits to small families like housing or education.
- Empower women through jobs, education, and property rights.

Link to Sustainable Development

Controlling population helps:

- Reduce pressure on nature and resources.
- Support fair and balanced economic growth.
- Improve people's lives and protect the environment.

Conclusion

If population growth is not controlled, it can damage the environment and society. Population control must be part of the larger plan for sustainable development to ensure a good future for all.

Crazy Consumerism

Crazy consumerism means buying and using goods much more than needed. It's often influenced by advertising, peer pressure, and the desire to show off wealth or status.

Key Features

- Buying luxury or unnecessary products.
- Short life of products—designed to break or go out of fashion.
- Culture of throwing away instead of reusing.
- Shopping based on trends or emotions.
- Choosing brand names over real usefulness.
- Thinking that success means owning more.

Environmental Effects

- Overuse of resources like water, fuel, and minerals.
- More garbage—plastic, electronic waste, packaging.
- Loss of forests and wildlife.
- More greenhouse gases leading to climate change.
- Pollution from factories and waste.

Social and Economic Effects

- Rich people consume more, poor people still struggle.
- People go into debt to buy more.
- Loss of traditional values and simple lifestyles.
- Developing countries follow harmful Western habits.

Causes

- Strong marketing and ads.
- Western lifestyle becoming global.
- Easy loans and online shopping.
- People feel pressure to keep up with others.
- Lack of awareness about environmental harm.

Control methods

- Buy only what is needed – practice conscious consumption.
- Repair, reuse, and recycle things.
- Support eco-friendly and ethical brands.
- Teach people about the environment.
- Set rules to control waste and false advertising.
- Focus on happiness and well-being, not just wealth.

Connection with Sustainable Development

Crazy consumerism works against sustainability goals:

- **SDG 12:** Use and produce responsibly.
- **SDG 13:** Fight climate change.
- **Other SDGs:** Protect nature and use resources fairly.

To protect the planet, we need to move from wasteful habits to mindful living.

Conclusion

Over-consumption harms the environment and society. We must change our behavior, values, and policies to support sustainable and fair lifestyles.

Environmental Education (EE)

Environmental education teaches people about nature, environmental problems, and how human actions affect the Earth. It aims to build knowledge, responsibility, and action for a sustainable future.

UNESCO Definition:

EE is a lifelong process that helps people of all ages understand the environment and the challenges it faces.

Main Goals of Environmental Education

- Raise awareness about environmental issues.
- Teach how ecosystems work.
- Encourage people to care for the environment.
- Build problem-solving skills.
- Get communities involved in protecting nature.
- Create a sense of responsibility and sustainability.

What Does It Include

- **Awareness** of environmental issues.
- **Knowledge** about pollution, climate, biodiversity, etc.
- **Positive Attitudes** toward the environment.
- **Skills** to find and solve problems.
- **Participation** in cleanups, campaigns, and actions.

Different Ways to Learn

- **Formal:** In schools and colleges (NCERT, UGC).
- **Non-formal:** Through NGOs, media, community events.
- **Hands-on:** Field trips, clean-up drives, nature walks.

- **Interdisciplinary:** Mix with science, ethics, policy, and more.

Importance

- Encourages smart and eco-friendly decisions.
- Helps with conservation and sustainability.
- Increases public understanding of nature.
- Empowers communities to take action.
- Tackles global issues like pollution, deforestation, water shortage, etc.

Role in EIA and Environmental Policy

- Educates stakeholders about environmental risks.
- Increases public involvement in EIA processes.
- Supports clear communication and informed consent.
- Helps in carrying out Environmental Management Plans (EMPs).

Challenges Faced

- Lack of trained teachers and good materials.
- Focus mostly on urban problems, not rural.
- Not given enough attention in school courses.
- Just giving knowledge isn't enough, behaviour change is needed.

Conclusion

Environmental Education is key to solving today's environmental problems. By including it at all levels of society, we can build a more informed, responsible, and sustainable future.

Urban Sprawl

Meaning

Urban sprawl means the spread of cities into nearby rural areas in an unplanned and scattered manner, usually along roads or highways. This leads to poor land use and harm to the environment.

Main Features

- Low population density and scattered buildings
- High use of private vehicles
- Different areas for homes, shops, and industries
- Encroachment on farmland and forests
- Lack of proper city planning and public transport

Why It happens

- Growing population and migration to cities
- Cheap land near city edges
- Poor planning and weak rules
- People want bigger houses
- More roads and transport access
- Land speculation by developers

Effects on the Environment

- Farmland and green areas lost
- Forests destroyed; wildlife affected
- More pollution due to vehicles
- Higher greenhouse gas emissions
- Depletion and pollution of water sources
- Blocked drainage systems causing floods

Social and Economic Effects

- Traffic jams and longer travel times
- Poor access to facilities for some people
- Pressure on infrastructure (water, roads, sewage)
- Poor communities may be left behind
- More public money needed for services

Urban Sprawl in EIA

Urban sprawl is studied in Environmental Impact Assessments (EIA) to:

- Check changes in land use

- See effects on farming and biodiversity
- Measure carrying capacity
- Plan for eco-friendly development

How to Control Urban Sprawl

- Smart city planning
- Building more in already developed areas
- Promoting public transport and walking/cycling
- Creating green belts to limit city spread
- Using land rules strictly
- Encouraging mixed-use buildings (homes + shops)

Conclusion

Uncontrolled urban sprawl is a serious issue. Planned development is needed to balance city growth and environmental care.

Human Health

Human health includes physical, mental, and social well-being. In environmental studies, it refers to how pollution and poor environments affect people's health.

Environmental Factors Affecting Health

- **Air pollution** – causes asthma, lung diseases
- **Water pollution** – leads to cholera, typhoid
- **Soil contamination** – harms through chemicals and pesticides
- **Noise pollution** – causes stress and sleep problems
- **Climate change** – increases diseases like dengue and malaria
- **Radiation** – can cause cancer or genetic issues

Workplace Health Hazards

- Exposure to chemicals, noise, and dust
- Poor posture and long hours causing strain
- Stress from tough working conditions
- Risk of accidents in risky jobs (mining, construction)

Health in EIA

- EIA includes Health Impact Assessment (HIA)
- HIA finds health risks early in a project
- It ensures preventive steps and fair health services
- Focuses on protecting vulnerable groups

Who Are Vulnerable

- Children and elderly
- Pregnant women
- Low-income communities
- People working outdoors or in slums

How to Protect Health

- Control pollution (air, water, soil)
- Proper waste disposal
- Regular health monitoring
- Health awareness campaigns
- Clean drinking water and hygiene
- Better hospitals and emergency services

Link to Sustainable Development

- Health is part of SDG 3: "Good Health and Well-Being"
- Clean environments support public health
- Sustainability helps future generations stay healthy

Conclusion

Protecting human health needs teamwork through better environment, healthcare, and education. It's also a major goal of sustainable development and environmental planning.

Environmental Ethics

Meaning

Environmental ethics is about what's right or wrong in our relationship with nature. It deals with our duty to care for the environment and other living beings.

Objectives

- Teach responsibility toward nature
- Show respect for animals and plants
- Set moral rules for saving nature
- Help in making good environmental decisions

Key Ideas

Anthropocentrism: Humans are most important; nature is for human use

Biocentrism: All life forms have value, not just humans

Ecocentrism: Whole ecosystems matter, including non-living parts

Deep Ecology: Humans should change their ways to protect nature

Stewardship: Humans should act as caretakers of the planet

Intergenerational Equity: Protecting nature for future generations

Why It Matters

- Helps use resources wisely
- Promotes animal and plant rights
- Encourages climate fairness
- Supports environmental laws and EIAs
- Balances development with conservation

Ethics in EIA

- Makes project planning more fair and responsible
- Ensures people have a say in decisions
- Uses principles like “precaution” and “polluter pays”
- Focuses on long-term impact, not just short-term gains

Common Ethical Issues

- Cutting forests for industries
- Wildlife trade and animal testing
- Global warming and its effects
- Dumping waste in poor communities
- Using fossil fuels instead of clean energy
- Rights of tribal and forest-dwelling people

Indian View

- Ancient texts show nature as sacred
- Respect for animals, rivers, and trees
- Gandhiji's idea: Live simply and in harmony with nature
- Indian Constitution (Article 51A(g)) asks all citizens to protect nature

8. Conclusion

Environmental ethics helps us make fair, long-term choices for nature and society. It guides laws, policies, and behaviour for a healthier planet.

Environmental Economics

Environmental economics studies how the environment and economy affect each other. It helps us understand the value of clean air, water, forests, and how to use resources efficiently.

Goals

- Put a value on nature and its services
- Study costs of pollution and resource loss
- Suggest smart tools for managing the environment
- Promote development that doesn't harm nature
- Support policy-making with economic logic

Main Concepts

Externalities: Hidden costs (like pollution) not in market prices

Market Failure: When markets don't work properly due to externalities

Natural Capital: Nature's stock of resources (like forests, rivers)

Polluter Pays Principle: Polluters must pay for the damage they cause

Cost-Benefit Analysis: Comparing project costs and benefits to the environment

Environmental Valuation: Giving a money value to things like clean air or beauty

Economic Tools

- **Taxes/Subsidies** : Carbon tax or rewards for green tech
- **Tradable Permits** : Trading pollution limits
- **PES** :Paying to protect forests or biodiversity
- **Green Accounting** : Adjusting GDP for environmental damage
- **EIA** : Includes cost-benefit and economic valuation

Use in EIA

- Estimate pollution costs
- Decide if a project's benefits are worth the harm
- Value lost forest, clean water, etc.
- Suggest eco-friendly alternatives
- Calculate compensation for affected people

Why It's Important

- Helps reduce waste and pollution
- Ensures fairness and justice in policies
- Gives solid data for decision-making
- Joins economics and environment for better planning

Challenges

- Hard to value nature correctly
- Poor data in developing countries
- Short-term growth vs. long-term sustainability

Conclusion

Environmental economics is essential for smart planning that respects both nature and economic needs. It leads to better, balanced development.

Green Building

A green building is one that is built and run in a way that uses fewer resources and causes less harm to the environment and people's health. It focuses on saving energy and water, using eco-friendly materials, and being sustainable.

Aims of Green Buildings

- Reduce damage to the environment
- Save water and energy
- Create healthier spaces to live and work in
- Encourage sustainable planning
- Use materials that are renewable or recyclable

Main Features

- Saves energy (e.g., solar power, LED lighting)
- Saves water (e.g., rainwater harvesting)
- Uses eco-friendly materials
- Reduces waste through recycling
- Has clean air and natural light inside
- Green roofs to keep buildings cool
- Modern and efficient heating/cooling systems

Advantages

- **Environment:** Less pollution, protect biodiversity
- **Economy:** Lower bills, better resale value, government support
- **Health:** Clean air inside, better comfort
- **Maintenance:** Easier and cheaper to maintain

Certifications

- **LEED:** Internationally used system
- **IGBC:** Indian standard for green buildings
- **GRIHA:** Developed in India (TERI + MNRE)
- **BREEAM:** UK-based rating system

Role in EIA

- EIA checks if the building plan is environment-friendly
- Encourages green design choices
- Identifies eco-friendly options
- Helps fight climate change

Government Support (India)

- **ECBC:** Promotes energy-saving buildings
- **Smart Cities Mission:** Focuses on green planning
- Many states offer rewards for green buildings

Summary

Green buildings are a smart choice for the future. They protect nature, save money, and keep people healthy—playing a key role in sustainable development.

Ecological Footprint

It is a measure of the amount of biologically productive land and water area required by an individual, population, or activity to produce the resources it consumes and absorbs the wastes it generates, using current technology and resource management practices. Ecological footprint is measured in global hectares (gha). It is a standardized unit representing the bio productive capacity of land.

Purpose

- Measure our demand on nature
- Check if our lifestyle is sustainable

- Find out if we are using more than Earth can handle
- Help in planning and saving resources

Main Parts

- **Carbon Footprint:** From burning fuel
- **Food Footprint:** Land needed for farming
- **Housing Footprint:** Space taken by buildings
- **Goods & Services:** Resources used for daily needs
- **Forests & Oceans:** Natural resources used

Bio capacity

- Earth's ability to recover and provide
- If demand exceeds supply that leads to Ecological Deficit
- If demand lesser than supply that leads to Ecological Reserve

Importance

- Shows how much pressure we put on nature
- Encourages us to reduce waste and overuse
- Helps governments make better plans
- Promotes conservation and balance

In EIA

- Checks long-term effects of projects
- Helps compare different project plans
- Points out areas that need improvement
- Supports low-impact and eco-friendly planning

Examples by Country

- **High footprint:** USA, UAE, Australia
- **Low footprint:** India, Bangladesh (but increasing)

Ways to Reduce It

- Use renewable energy
- Eat local, less meat-based food
- Cut down waste
- Use public transport
- Educate and make smart policies

Summary

Ecological footprint helps us understand our impact on the planet. It supports better living and smarter development choices.

Low Carbon Lifestyle

Living in a way that reduces carbon emissions . This means using less fuel and electricity, wasting less, and making daily habits more eco-friendly. It supports the fight against climate change.

Main Goals

- Lower greenhouse gas (GHG) emissions
- Save energy and resources
- Encourage green choices in life
- Support climate action (SDG 13)

Ways to adopt Low Carbon Life Style

Area	Actions
Energy	Use solar panels, LEDs, efficient devices
Transport	Walk, cycle, use public transport, electric vehicles
Food	Eat local food, waste less, more plant-based meals
Waste	Reuse, recycle, compost
Shopping	Buy less, choose durable goods
Water	Harvest rain, fix leaks, use low-flow taps
Digital Use	Avoid e-waste, reduce screen time

Benefits

- **Environment:** Less pollution
- **Money:** Lower bills
- **Health:** Healthier lifestyle
- **Society:** Greener, aware communities

Role in EIA

- Used as a method to reduce pollution from projects
- Helps plan green buildings and transport
- Supports low-carbon infrastructure
- Helps calculate and limit emissions in planning

Government Support (India)

- **LiFE Movement:** Promotes sustainable habits
- **UJALA Scheme:** Distribution of LED bulbs
- **PAT Scheme:** Improves industry energy use
- **EV Support:** Encourages electric vehicles
- **Solar Rooftop Schemes:** Clean power at home

Contribution of an individual

- Turn off lights when not needed
- Carry your own bag and bottle
- Say no to single-use plastics
- Learn and teach about climate change
- Buy from eco-friendly local sellers

Summary

A low carbon lifestyle is an easy but important way to fight climate change. Small steps taken by many people can lead to a big positive impact on the planet.

Recommended Books

1. **Canter, L.W.**
Environmental Impact Assessment
(McGraw-Hill Education)
Classic textbook covering EIA methods and environmental planning.
2. **Rau, G.J. and Wooten, D.C.**
Environmental Impact Analysis Handbook
(McGraw-Hill)
Detailed guide to EIA processes and practical approaches.
3. **R.K. Khitoliya**
Environmental Pollution: Management and Control for Sustainable Development
(S. Chand Publishing)
Good Indian textbook with applied aspects of EIA and sustainability.
4. **R.B. Singh**
Ecology, Environmental Science and Conservation
(S. Chand & Co.)
Covers conservation, pollution, ecological footprint, and sustainable practices.
5. **Kibert, Charles J.**
Sustainable Construction: Green Building Design and Delivery
(Wiley India)
Comprehensive guide to green building techniques and certification.
6. **Jagadish, K.S.**
Alternative Building Materials and Technologies
(New Age International Publishers)
Focuses on eco-friendly construction in the Indian context.
7. **Satish Kumar and Vibhuti Garg**
Energy Efficiency in Buildings in India
(TERI Press)
Discusses ECBC, energy-saving designs, and green certification in India.
8. **Singh, Savindra**
Environmental Geography

(Prayag Pustak Bhawan, Allahabad)

Includes topics on carbon lifestyle, human-environment interaction.

9. **Botkin, D.B. and Keller, E.A.**

Environmental Science: Earth as a Living Planet

(Wiley)

Covers global sustainability issues and ecological footprints.

10. **Yashpal Sharma and M.P. Joshi**

Fundamentals of Environmental Studies

(Lakshmi Publications)

UG-level text, includes sustainable development and low-carbon living.